Question-conditioned image-language training

a Observed training images: two questions about one paragraph

Question region Q0

哪个人是世纪越南反唐起义军的军事首领？

Recorded answer Y0 冯兴

Question region Q1

冯兴与他的弟弟冯骇接受谁的建议，率众起义？

Recorded answer Y1 杜英翰

Each full prompt Xi also contains the same paragraph image. Appended white termination blocks are not drawn.

b Four causal likelihoods under the same parameters

(X0, Y0) (X0, Y1)
(X1, Y0) (X1, Y1)

→

Shared image generator
490M parameters, 8 experts

→

	Y0	Y1
X0	e_{00}	e_{01}
X1	e_{10}	e_{11}

eij = mean negative log likelihood of the complete answer suffix, including termination

c Positive image likelihood with paired question assignment

$$D = e_{01} + e_{10} - e_{00} - e_{11}$$

λ = 0: control; λ = 1: assignment

$$L = (e_{00} + e_{11}) / 2 + \lambda \cdot 0.01 \cdot \mathrm{softplus}(-D / 0.01)$$

Observed targets, not model outputs. Both arms start from the same V62 EMA; autonomous answers are evaluated separately.